

English in Arabic Script

Notation, Evidence and Tools

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Abstract

This monograph specifies a writing system in which ordinary English is written, word for word and without translation, in standard Arabic letters with optional short vowel marks. It reconstructs the system from the surviving evidence: a handwritten page, the typed passage it copies, and several generated test texts. It records which conventions the evidence settles, which it leaves open, and exposes each open point as a named option in a Python toolkit. The toolkit contains a forward transliterator driven by the CMU Pronouncing Dictionary, a reverse reader that returns ranked candidates, an inventory linter, a corpus statistics tool, and a glossary generator. Every table and figure of agreement in the text is produced by the code and the specimens, so the numbers cannot drift from the implementation.

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Chapter 1

Scope and Definition

1.1 What is being written

The system writes English. Vocabulary, word order, sentence boundaries and punctuation stay English; only the letters change. A reader who sounds out the Arabic letters recovers the sound of the English word, and a reader who knows English recovers its meaning. Nothing is translated. The sentence

إِنْ ذِسْ بَارْتُ أَفْ هِيلِنْ رُولِنْزْ سَايْكُوسِينَمَا

means *In this part of Helen Rollins' psychocinema* because it sounds like that English phrase, not because any word in it is an Arabic word.

The intended product is a book written in this orthography, accompanied by a pronunciation key and parallel examples. The tools described here exist to produce and check such a book, not to replace the decisions about how the orthography should look.

1.2 Constraints

Four constraints recur in the source material and define the system.

1. **English throughout.** Earlier generated texts repeatedly drifted into Arabic vocabulary, for instance spelling out numbers as Arabic number words or substituting a real Arabic word where an English word was expected. Each such drift is an error against this constraint.
2. **Standard Arabic letters only.** The Persian letters پ, ف, گ, چ, ژ are excluded. English sounds that Arabic lacks are approximated with standard letters.
3. **Short vowels are optional marks.** Fatha, damma, kasra and sukoon may be written to help the reader. Both fully marked and unmarked presentations were requested, so mark density is a presentation setting, not part of the spelling.
4. **Approximation is acknowledged.** Several English sounds share a letter, and several English vowels share a representation. The orthography is an approximate phonetic notation. Claims that it reproduces English pronunciation exactly are not made here.

1.3 What is not claimed

Interpretive frameworks attached to the project in earlier discussions, including semiotic or epistemological readings, are outside this monograph. They would need separate justification before becoming part of the foundation. The keyboard layout used for typing the script is treated as an input method, distinct from the sound key (Chapter 12).

1.4 Organisation

Part one of the work (Chapters 2 to 5) reconstructs the notation from evidence. Chapters 6 to 10 describe the tools and how they are checked. Chapters 11 to 13 cover teaching formats, typesetting and open work. The appendices give the command line, the full list of attested forms and the legacy sources.

1.5 What the notation is for

The notation serves three uses that pull in slightly different directions. As a reading aid, it lets a reader of Arabic script sound out English text without any knowledge of English spelling, because every word is written as it is said. As a writing system in its own right, it allows English to be handwritten quickly with a small inventory of letters and marks, which is how the handwritten page was produced. As a typographic exercise, it asks how far a script built for another language can carry English before it breaks. The tools in this book support all three. They differ only in how many marks are shown, which is why mark density is a presentation setting and not a change of notation.

1.6 Design principles

Six principles run through the later chapters. They are stated here once so that individual rules can be judged against them.

1. **English words, English order.** Nothing is translated and nothing is reordered. A line of the notation has the same number of words as the English line it comes from, and a reader can align the two word by word.
2. **Standard Arabic letters only.** The letters of the basic Arabic block are used, together with fatha, damma, kasra and sukoon. Persian letters such as پ, چ, گ and ف are excluded, and near equivalents are accepted instead.
3. **Near equivalents over new letters.** Where English has a sound that Arabic lacks, the nearest Arabic letter is used. English *p* is written ب and English *v* is written ف, as a voiced *f*. Exact recovery of short words is not required.
4. **Marks are optional.** The same letters can be shown with every mark, with vowel marks only, or with none, and the choice never alters a letter.
5. **Evidence before invention.** A rule is added only when specimens support it, and each unsupported choice is a switch with a stated default.
6. **Ambiguity is reported.** The notation merges some distinctions, so the reverse tool returns ranked candidates and does not pretend to recover spelling exactly.

1.7 How to approach the text

A reader of this notation is reading English. The Arabic letters are a set of learned conventions for writing it, and they do not turn the text into Arabic or into a transliteration to be decoded each time. At first, sounding out every letter is the natural strategy, and it works because the notation is written as the words are said. With practice the strategy fades. Familiar words become recognisable as whole shapes, as printed English words do, and frequent words such as *دَا* (*the*) and *أَنْدُ* (*and*) are recognised before their letters are consciously read. Letter-by-letter decoding is therefore a stage in learning, and it is not the reading experience the notation aims at. Later chapters that give a Latin respelling beside the Arabic-script line offer it as a support for that early stage.

1.8 A first reading

The sentences below were composed in the notation and then checked against the toolkit. Each block gives the Arabic-script form with all marks, a plain Latin respelling that is only a reading aid, and the English original. The second line of the first two blocks shows the same words with all marks removed, which is how the notation looks at its lightest.

دَا بُوكُ إِزْ رِتِنْ إِنْ إِنْغَلِشْ.
 دَا بوكُ إِزْ رتنْ إِنْ إِنْغَلِشْ.
dhu buk iz ritun in ingglish
 The book is written in English.

دَا لَتَرَزْ تَشِينْجْ، بَتْ دَا وَرْدَزْ رَمِينْ.
 دَا لترز تشينج، بت دَا وردز رمين.
dhu leterz cheynj but dhu werdz rimeyn
 The letters change, but the words remain.

أَ رِيدَرْ لَرَنْزْ دَا سَكْرِبَتْ أَنْدُ يَغْنَزْ تُو رِيكُغَنَائِزْ أَ فَمِيلِيرْ فُويسْ.
 أَ ريدر لرنز دَا سكربت أند بغنز تو ريكغنايز أَ فميليير فويس.
u reeder lernz dhu skript und biginz too rekugnayz u fumilyer voys
 A reader learns the script and begins to recognise a familiar voice.

Several choices of the system are visible even in these three sentences. English *p* in *script* is written ب. English *v* in *voice* is written ف. The *g* of *English* and *begins* is written غ, and *ch* in *change* is the two-letter sequence ش. The word *book* keeps a waw because English writes two letters for its vowel, a point that Chapter 5 treats as a general principle.

1.9 Plan of the book

Chapters 2 and 3 describe the specimens and separate what they establish from what remains open. Chapters 4 and 5 state the sound key and the spelling-conditioned vowel rules. Chapters 6 and 7 describe the software architecture and the forward transliterator, and Chapter 8 the reverse reader. Chapters 9 and 10 cover checking and evaluation. Chapter 11 collects worked passages, Chapter 12 deals with typesetting and input, and Chapter 13 lists the open questions. The appendices give the command-line reference, the attested forms and the legacy sources.

Chapter 2

Specimens and Provenance

2.1 Evidence tiers

Not all specimens carry equal authority. The reconstruction ranks them.

Tier 1: the handwritten page. A page written by hand in the orthography, later transcribed with every mark. It contains four lines:

كَاتِبَالِسْتُ يُثَوِّبَانِيْرْمُ
دَا رِبْرِشْنُ أَفْ لَاكْ أَنْدَ إِنْسَنُ كَلْتَشَرْلُ سِمْبُئْمُزْ.
إِنْ دِسَنُ بَارْتْ أَفْ هِيلِنْ رُولَنْزْ سَايْكُوسِيْنِمَا،
شِي دِلْفَرْزْ إِنْتُو دَا تَيْتَشَرْ أَفْ لَاكْ.

It is the only specimen produced by the author's own hand and is treated as authoritative.

Tier 2: the typed passage. A longer text of which the page is a copy of the opening. It was produced by an assistant working under the author's instructions and accepted as good, with the explicit acknowledgement of glitches and inconsistencies. Acceptance of a specimen did not certify every spelling in it.

Tier 3: the glossary passage. An earlier passage with a line-by-line glossary. Its commentary contradicts its examples: it announces one treatment of *p* and *v* while the examples show another. Examples outrank commentary.

Tier 4: generated test texts. Summaries of *Micromegas* and *Atlas Shrugged*, transliterated by an assistant. The *Atlas* text uses four Persian letters; the *Micromegas* text uses one.

Unusable: the chat export. The original conversation titled *English in Arabic Text* survives only as a text export in which every Arabic character has been replaced by the replacement character U+FFFD. It documents the instructions given, but supplies no letter-level evidence.

2.2 Inventory of the specimens

Table 2.1 counts letters and marks per specimen using the corpus tool (`arabglish stats`) and the linter (`arabglish lint`). The page and typed passage use only standard letters. The *Atlas* text accounts for 125 occurrences of letters outside the standard inventory.

Specimen	Letters	Marks	Extended letters	Lint issues
Page transcript	99	80	none	0
Typed passage	954	791	none	0
Glossary passage	711	545	none	3
Micromegas	1417	364	ف 71	21
Atlas Shrugged	1422	922	ف 75 گ 03 پ 43 ج 4	125

Table 2.1: Letter, mark and inventory statistics for each specimen. Extended letters are letters outside the standard Arabic block, with their counts.

2.3 Legacy tools

Two earlier tools exist and are preserved verbatim in the project under `legacy/`.

The keyboard layout. An AutoHotkey script maps Latin keys to Arabic letters one key at a time. Its assignments, such as the key *p* producing a two-letter sequence, are key positions chosen for typing. They record how to enter characters, not how English sounds are written, and earlier assistant commentary that read them as sound correspondences conflated the two.

The Python script. A pronunciation-driven script uses the CMU Pronouncing Dictionary to obtain English phonemes and maps them to Arabic characters. The preserved copy is the revision with Persian letters removed, as it appears in a pasted conversation, so it may differ from the file that was run. Its fixed word table (*the, this, that, of, and, in, to, a, is*) is adopted in the toolkit. Its structural defects are listed in Appendix C.

2.4 How the specimens were transcribed

Three of the specimens are the author’s own writing. The handwritten page was photographed and typed out letter by letter with every mark, so that the transcript preserves the author’s vowel marks and sukoon. The typed passage and the glossary passage were already in digital form. The generated texts, which the author produced with language-model help in earlier sessions, were taken as found.

Transcription from the photograph has two hazards. Fatha, damma and kasra are small and can be confused with dots and with each other, and a sukoon above a letter can look like a damma. Wherever the photograph was ambiguous the typed transcript supplied by the author was treated as authoritative. The transcript is stored in `specimens/page_handwritten_transcript.txt` and its four lines are aligned with the English by `scripts/build_attested.py`, which stops with an error if the token counts disagree.

2.5 Alignment of Arabic tokens to English words

Every comparison in this book depends on knowing which Arabic token stands for which English word. For the page and the typed passage the author supplied the English text, and tokens were paired in order. The script asserts that the number of Arabic tokens equals the number of English words in each paragraph, so an insertion or omission anywhere in the text is caught instead of silently shifting every later pair. The result is the list of 45 word forms in Appendix B.

2.6 Strengths and weaknesses of each specimen

Handwritten page. Closest to the author’s practice and fully marked, but only four short lines.

Typed passage. About one thousand letters of fully marked text in the author's conventions. It is the main source of attested forms. It was produced partly by typing and partly with assistance, so it carries some noise, which Chapter 5 notes where it matters.

Glossary passage. Marks and consonants are consistent, but its own commentary contradicts its letters for the choice of *v*. Its letters were weighted above its commentary.

Micromegas and Atlas. Early generated texts. They show the system at an earlier stage and with Persian letters in the Atlas text. They are used for the inventory counts and the linter, and not for attested forms.

2.7 What the specimens cannot show

No specimen was written under controlled conditions. The author knew the conventions while writing, and the specimens were not produced from dictated or unseen text. They therefore show how the author writes words the author already has in mind, and give little information about how the author would write a word met for the first time. This limitation returns in Chapter 10, where agreement figures are discussed, and in Chapter 13, where held-out specimens are the first request.

Chapter 3

Evidence and Open Decisions

3.1 Consonants

Table 3.1 gives the consonant key as implemented, with a status for each sound. *Firm* means the specimens agree. *Open* means they conflict. *Proposed* means the author has named a preference that later generated passages did not follow. *Provisional* means no specimen word settles it.

ARPAbet	Example	Letters	Status	Note
P	part	ب	firm	ب in every specimen; no specimen uses peh
B	big	ب	firm	
T	top	ت	firm	
D	dog	د	firm	
K	cat	ك	firm	غ proposed by the author; ج appears in generated passages
G	gap	غ	proposed	
F	fan	ف	firm	ف by default (v is a voiced f); ب in universal, individual; see census
V	very	ف	open	
TH	think	ث	firm	
DH	this	ذ	firm	
S	sun	س	firm	ز in the cleaned script; no specimen word
Z	zoo	ز	firm	
SH	she	ش	firm	
ZH	vision	ز	provisional	
HH	hat	ه	firm	ش on the handwritten page; ش in one typed word
CH	nature	تش	firm	
JH	subject	ج	firm	ج
M	man	م	firm	ج follows G; typed passages show ج
N	no	ن	firm	
NG	sing	نغ	provisional	ج follows G; typed passages show ج
L	lack	ل	firm	
R	red	ر	firm	و as a consonant
W	way	و	firm	
Y	yes	ي	firm	ي as a consonant

Table 3.1: Consonant key (default profile). English *p* is written ب in every specimen.

3.2 English v

The evidence on *v* is divided. Table 3.2 lists every *v* word that could be checked in the specimens. The letter **ب** appears 7 times and **ف** appears 5 times. The author’s own page writes *delves* with **ف**. The typed passage writes *universal*, *individual*, *however*, *generative*, *never*, *resolved* and *deliver* with **ب**. The earlier glossary passage writes *subjective*, *living*, *reveal* and *selves* with **ف**, while its commentary says the opposite.

English	Spelling	Letter for v	Specimen
delves	دَلْفَزْ	ف	page, typed
universal	بُونِيْرَسَلْ	ب	typed
individual	اِنْدِيْدِيْوَالْ	ب	typed
however	هَوِيْرْ	ب	typed
generative	جِيْرَاتِبْ	ب	typed
never	نِيْرْ	ب	typed
resolved	رَزُولِنْدْ	ب	typed
deliver	دَلِيْرْ	ب	typed
subjective	سُوْبِيْجِكْتِيْفْ	ف	glossary
living	لَاْفِيْجْ	ف	glossary
reveal	رِيْفِيْلْ	ف	glossary
selves	اِنْسَلِيْفَزْ	ف	glossary

Table 3.2: Census of English *v* in the specimens that could be aligned.

The earlier Python script used a voicing heuristic, choosing **ب** between voiced neighbours and **ف** otherwise. That heuristic cannot reproduce the census: *delves* and *resolved* have identical neighbourhoods (a liquid before *v*, a voiced consonant after) yet receive different letters. The toolkit therefore offers three policies, `b`, `f` and `context`, and a per-word override through the attested list. The default is **ف**, on the author’s reasoning that *v* is a voiced *f*; this also matches the old program and the page’s *delves*. The typed passage’s *individual* and *universal* use **ب**, and `--v b` reproduces that choice.

3.3 Hard g and the velar nasal

The author proposed غ (ghain) for hard *g*. Generated passages used ج (jeem), and the cleaned Python script mapped *g* to ج, which merges *g* with *j* in *subject*. The default is غ, with `--g jeem` available. The velar nasal follows *g*: نغ by default and نج under `jeem`.

3.4 The affricate ch

The page writes *cultural* and *nature* with the two-letter sequence تشْ, taking sukoon on the first letter. One typed word, *nature*, uses شْ alone. The sequence is the default. When the sequence closes a word in full marks both letters carry sukoon, as in وُتْشْ *which*.

3.5 Remaining decisions

Table 3.3 lists the decisions and where the toolkit exposes them.

Decision	Default	Switch
English <i>v</i>	ف	--v b f context
Hard <i>g</i>	غ	--g ghain jeem
Persian letters	excluded	--persian (for conversion only)
Mark density	full marks	--density full medial vowels none
<i>the</i> before a vowel	لّ always	--the-before-vowel
Numbers	Arabic-Indic digits	--digits indic latin spoken
Specimen forms	computed forms	--attested
Doubled letters	not marked (no shadda)	none

Table 3.3: Open decisions and their switches.

3.6 Decision log

The consonant and vowel choices are recorded here with the evidence and reasoning behind each, in the order in which they were settled.

English p. Written ب in every specimen and in the old program. The author states that ب for *p* is acceptable and that exact recovery of short words is not required. Settled.

English v. The old program wrote ف. The typed passage writes ب in *universal* and *individual* and the page writes ف in *delves*. The author's reasoning is that *v* is only a voiced *f*, so the default is ف. The alternative is kept as --v b.

Hard g. The old program wrote ج, merging it with *j*. The author later proposed غ, which keeps *g* distinct from *j*. The default is غ and --g jeem restores the older choice.

English ch. The page writes the two-letter sequence ئش, the typed passage writes ش in *nature*, and the old program wrote ش. The sequence is the default because it keeps *ch* separate from *sh*.

English ng. Written as nasal plus ghain, نغ. The old program wrote ن alone. A following hard *g*, as in *English* and *finger*, is not written twice.

Voiced th and voiceless th. Written ذ and ث in every source.

Zh. Written ز. There is no better standard Arabic letter and no specimen word with the sound is attested.

Vowel colour of unstressed syllables. Taken from the vowel letter of the spelling, with the exceptions listed in Chapter 5.

3.7 Consonant examples

The following words exercise the consonant decisions: *thing*, *think*, *this*, *that*, *church*, *judge*, *measure*, *vision*, *singer* and *finger*, with a few clusters.

3.8 Decisions that need no more evidence

Not every open choice is equally pressing. The choice for *v* affects a small group of words and is a single switch. The choice for hard *g* affects more words but the two options are equally readable. What matters most for the next stage is a larger set of fully marked words in the author's own hand, because most of the vowel rules rest on a few dozen forms.

English	Arabic script	Respelling
thing	شَيْءٌ	<i>thing</i>
think	يَتَفَكَّرُ	<i>think</i>
this	ذَٰلِكَ	<i>dhis</i>
that	ذَٰلِكَ	<i>dhat</i>
church	كَنِيسَةٌ	<i>cherch</i>
change	تَسْيِيرٌ	<i>cheynj</i>
judge	حَاجٌّ	<i>juj</i>
measure	مِيزَرٌ	<i>mezher</i>
vision	فِيْزْنٌ	<i>vizhun</i>
singer	سِنْعَرٌ	<i>singer</i>
finger	فِنْغَرٌ	<i>fingger</i>
garden	غَارْدُنٌ	<i>gaardun</i>

Table 3.4: Words that exercise the consonant decisions.

Chapter 4

The Sound Key

4.1 Vowels

Each English vowel is written as a short mark on the preceding consonant and, for long vowels and glides, a vowel letter. Table 4.1 shows the default correspondences using words from the page. Phonemes are named in ARPAbet, the notation of the pronouncing dictionary.

ARPAbet	Mark	Letters	Example	Rendered
AA	fatha	ا	part	بَارِطْ
AE	fatha	–	lack	لَاكْ
AH	fatha	–	cultural	كُلْتُشَرْلْ
AO	damma	و	on	أُونْ
AW	fatha	او	flower	فُلَاوَزْ
AY	fatha	اي	psychocinema	سَايْكُوْسِيْنِمَا
EH	kasra	–	delves	دَلْفَزْ
ER	fatha	ر	nature	نَيْشَرْ
EY	fatha	ي	nature	نَيْشَرْ
IH	kasra	–	this	زِسْ
IY	kasra	ي	she	شِي
OW	damma	و	Rollins	رُولِنْزْ
OY	damma	وي	void	فُوَيْدْ
UH	damma	–	could	كُوْدْ
UW	damma	و	into	إِنْتُو

Table 4.1: Vowel key. The mark is the short vowel; the letters are the vowel letters added for long vowels and glides.

Three kinds of vowel letter occur. A *long* letter, as in سِيْنِمَا, takes no sukoon. A *glide* letter, the second element of a diphthong as in سَايْكُو, takes sukoon. A *rhotic* vowel is written as a mark followed by ر, as in شَرْ.

4.2 Sukoon

Sukoon marks a consonant followed by no vowel. In full density every such consonant carries it, including the last consonant of a word, as in لَاكْ *lack* and كَابِيَالِسْتْ *capitalist*. Sukoon records the absence of a

vowel; it does not make a letter silent, and English double letters are not marked with shadda.

4.3 Word-initial vowels

A word that begins with a vowel takes a hamza seat: **أ** for fatha and damma, **إ** for kasra, as in **أف** *of* and **إن** *in*. An initial diphthong beginning with fatha and alef is written with **آ**. A vowel that follows another vowel is hosted by a preceding **ي** when there is one (**ئي**) and otherwise takes a seat.

4.4 Frequent words

Frequent function words have fixed spellings (Table 4.2). These come from the original fixed word table and from forms that the page and the typed passage share. They also resolve the contextual variation of *the* and *a*: the toolkit writes **لَا** before a vowel too, as the typed passage does, and offers the phonemic alternative **يَ** as an option.

English	Spelling	English	Spelling
the	لَا	is	إِزْ
this	ذِسْ	she	شِي
that	ذَتْ	it	إَتْ
of	أَفْ	its	إِسْ
and	أَنْدْ	as	أَرْ
in	إِنْ	into	إِنْتُو
to	تُو	with	وَدْ
a	أَ	are	أَرْ

Table 4.2: Fixed spellings of frequent words.

4.5 Punctuation and numbers

The comma, semicolon and question mark take their Arabic forms **،**, **؛**, **؟**. A possessive apostrophe after *s* is silent, as on the page. Digits default to Arabic-Indic numerals. A `spoken` option writes numbers as English number words in the notation (*twenty five*), which avoids the earlier failure of rendering numbers as Arabic number words.

4.6 Letter sound and assigned function

Each Arabic letter has an ordinary sound in Arabic, and in this notation it also has an assigned function in writing English. The two often coincide, as with **ب** for *b* and **ل** for *l*, and sometimes they differ. The letter **غ** is a voiced uvular fricative in Arabic, and it is used here for the hard English *g* by convention. The same holds for **ث** and **ذ**, which are close to English *th* and *dh* sounds, and for **پ**, which stands for both *b* and *p*. The letters **و** and **ي** serve as consonants (*w* and *y*) and also as vowel letters.

Knowing Arabic script is therefore a useful starting point, since the letter shapes, the direction of writing and the role of the marks are already familiar. It does not supply every reading rule. A reader who pronounces **غ** as in Arabic will mispronounce *good* and *begin*, and a reader who expects **ب** to be only *b*

will miss the *p* in *script*. The tables in this chapter give the assigned function, and the Arabic sound is only a mnemonic where it helps.

4.7 Short vowels in practice

Short vowels are written with a single mark on the preceding consonant and no vowel letter. The words below show the five English short vowels, each followed by one consonant or a cluster.

English	Arabic script	Respelling
cat	كَاتْ	<i>kat</i>
bed	بِدْ	<i>bed</i>
sit	سِيَتْ	<i>sit</i>
hot	هُوَتْ	<i>haat</i>
cut	كَتْ	<i>kut</i>
put	بُتْ	<i>put</i>
pen	يِنْ	<i>pen</i>
map	مَاْبْ	<i>map</i>
top	تُوْبْ	<i>taap</i>
sun	سِنْ	<i>sun</i>

Table 4.3: Short vowels, written with a mark only.

4.8 Long vowels and the final e

English marks long vowels in several ways: a doubled letter, a vowel digraph, or a final silent *e*. The dictionary reading already gives the long vowel in each case, so the notation writes a vowel letter after the mark. The final *e* itself is silent and is not written.

English	Arabic script	Respelling
make	مَيِّكْ	<i>meyk</i>
time	تَائِمْ	<i>taym</i>
home	هُوْمْ	<i>hohm</i>
use	يُوسْ	<i>yoos</i>
cake	كَيِّكْ	<i>keyk</i>
like	لَايِّكْ	<i>layk</i>
phone	فُوْنْ	<i>fohn</i>
tune	تُوْنْ	<i>toon</i>

Table 4.4: Long vowels and glides written for the silent-e pattern.

4.9 Doubled letters and vowel digraphs

When the English spelling uses two vowel letters for one vowel, the notation keeps a waw or a yeh. This applies to the short vowel of *book* as much as to the long vowel of *food*. The spelling carries the weight here and the Arabic letter signals it. Chapter 5 explains why *book* is بُوكْ and not بُكْ.

English	Arabic script	Respelling
book	بُوكُ	<i>buk</i>
look	لُوكُ	<i>luk</i>
good	عُودُ	<i>gud</i>
foot	فُوتُ	<i>fut</i>
wood	وُودُ	<i>wud</i>
food	فُودُ	<i>food</i>
moon	مُونُ	<i>moon</i>
school	سُكُولُ	<i>skool</i>
too	تُو	<i>too</i>
two	تُو	<i>too</i>
blue	بُلُو	<i>bluo</i>
true	تُرُو	<i>troo</i>

Table 4.5: Words with a written double o, and related spellings.

English	Arabic script	Respelling
see	سِي	<i>see</i>
tree	تِرِي	<i>tree</i>
green	عَرِيْن	<i>green</i>
sleep	سَلِيْب	<i>sleep</i>
head	هَدْ	<i>hed</i>
bread	بَرْدْ	<i>bred</i>
read	رَدْ	<i>red</i>
meet	مِيْتْ	<i>meet</i>
meat	مِيْتْ	<i>meet</i>
dream	دَرِيْمْ	<i>dreem</i>
dead	دَدْ	<i>ded</i>

Table 4.6: Words with ee and ea.

4.10 Glides and diphthongs

The diphthongs of English are written as a vowel mark followed by a glide letter that takes sukoon, so *day* is دَـيْ, *light* is لَـيْتْ and *voice* is فُـوَيْسْ. This is the placement seen on the handwritten page.

4.11 R-coloured vowels

A vowel followed by *r* in the dictionary is written as a vowel mark and ر. The mark of an unstressed final syllable is fatha in every attested form, as in تَيِّنَسْر and هَوَيَّرْ. Stressed syllables take their mark from the spelling.

4.12 Clusters and sukoon

English has many consonant clusters that Arabic does not allow at the start of a word. The notation writes them with sukoon on all but the last consonant, so the cluster *str* in *street* appears as سْتَرِيْتْ. Word-final consonants also take sukoon in the full density. The medial density removes the final sukoon, which leaves clusters inside words intact.

English	Arabic script	Respelling
rain	رَيْنْ	<i>reyn</i>
day	دَيْ	<i>dey</i>
light	لَايْتْ	<i>layt</i>
night	نَايْتْ	<i>nayt</i>
voice	فُؤَيْسْ	<i>voys</i>
flower	فَلَاوَزْ	<i>flower</i>
hour	أَوْزْ	<i>ower</i>
boy	بُؤَيْ	<i>boy</i>
toy	تُؤَيْ	<i>toy</i>
noise	نُؤَيْرْ	<i>noyz</i>
now	تَاوْ	<i>now</i>
how	هَاوْ	<i>how</i>

Table 4.7: Glides and diphthongs.

English	Arabic script	Respelling
car	كَارْ	<i>kaar</i>
far	فَارْ	<i>faar</i>
her	هَرْ	<i>her</i>
bird	بِرْدْ	<i>berd</i>
turn	تَرْنْ	<i>tern</i>
word	وَرْدْ	<i>werd</i>
door	دَوْرْ	<i>dawr</i>
more	مَوْرْ	<i>mawr</i>
fire	فَايْرْ	<i>fayer</i>
tire	تَايْرْ	<i>tayer</i>

Table 4.8: R-coloured vowels.

English	Arabic script	Respelling
string	سْتَرِيْعْ	<i>string</i>
spring	سَبْرِيْعْ	<i>spring</i>
splash	سَبْلَاشْ	<i>splash</i>
street	سْتَرِيْطْ	<i>street</i>
strength	سْتَرِيْعَكْثْ	<i>strengkth</i>
twelfth	تَوَلْفَتْ	<i>twelfth</i>
asked	أَسْكُتْ	<i>askt</i>
texts	تَكْسُتْسْ	<i>teksts</i>

Table 4.9: Clusters, written with sukoon.

Chapter 5

Vowels Conditioned by Spelling

5.1 Observation

A purely phonemic mapping from the pronouncing dictionary reproduces the page's consonants well and its vowels less well. Comparing the computed forms with the attested forms shows a regular departure: the letter used for a vowel sometimes follows the English spelling rather than the sound.

1. **Stressed o.** The dictionary gives *Rollins*, *on*, *psychology* and *concept* an open back vowel. The specimens write them with damma and waw: رُولِنَزْ, أَوْنُ, سَايْكُولُوجِي, كُونْسِبَبْتُ. Contrast *part*, written بَارْتُ with fatha and alef, where the spelling letter is *a*.
2. **Open-syllable e and i.** *Helen* and *psychocinema* are written with kasra and هِيلِنُ ي: سِيْنِمَا. The stressed vowel letter is followed by one consonant and another vowel. In *delves*, *this* and *its*, where the vowel letter is followed by a cluster, only kasra is written.
3. **Schwa colour.** An unstressed vowel takes the colour of its letter: *a* gives fatha, *e* and *i* give kasra, *o* gives damma, as in سِيْمَبْتُمْزُ *symptoms*.
4. **Short initial vowels.** An unstressed vowel in the first syllable is written short, with no vowel letter: رِبْرَشْنُ *repression*, يُتُوبِيَانِيْزْمُ *utopianism*.
5. **Hosted hiatus.** A vowel after ي is written on the ي, as in يَي and ثِي.

5.2 Why pronunciation and spelling both contribute

The notation could have been defined as a pure phonetic transcription, with each vowel written from its dictionary sound alone. The author's practice shows that it is not. Familiar English words carry a visual identity that readers use in recognising them, and the notation keeps some of it. The word *book* is written بُوْكَ with a waw because English writes two vowel letters, and the waw retains that feature even though it does not reproduce the short vowel narrowly. The waw is a deliberate orthographic choice. It is not an error of pronunciation to be corrected.

The consequence for later revision is practical. A departure from dictionary pronunciation in an attested form is not automatically a mistake. It may be the spelling showing through, and a rule that removes it would make the notation more phonetic and less like the author's writing. Before such a form is changed, the question to ask is whether the departure is systematic across words with the same spelling, as the waw of *book*, *good* and *foot* is, or whether it is a lapse.

5.3 Reading of the pattern

These observations suggest that the notation is a hybrid. It is phonetic in its consonants and in the broad shape of its syllables, and orthographic in the choice of some vowel letters. The evidence is one author's page, a typed passage and generated texts, so the hybrid reading is a hypothesis about how the notation behaves, not a finding about how it must behave. The rules implemented from it are listed with their results in Chapter 10.

5.4 Remaining disagreements

After the rules are applied, 10 of the 45 attested forms still differ from the computed forms by more than marks. Table 5.1 lists them. They fall into five groups:

Lexical alef. *capitalist*, *capitalism*, *utopianism* and *psychoanalytic* carry an alef where the rules give only fatha. Other words with the same spelling pattern, such as *fundamental* and *cultural*, do not. No rule from spelling or stress separates them, so these stay in the attested list.

The choice of v. *individual* and *universal*, written with ب in the typed passage, discussed in Chapter 3. The default ف is the author's decision and the specimens disagree with it in these two words.

Failed alignment. When the number of vowel letter groups differs from the number of vowel phonemes, as in *ideology* and *individual*, spelling-based colouring is skipped and the phonemic default applies.

Seat letter. The article *a* is written آ in the typed passage and ا in the original fixed table. Both appear in the sources; the toolkit keeps the hamza form.

Generated-text noise. *nature* in the typed passage, which uses ش where the page has ثش, and *structures*, whose typed spelling departs from both the page conventions and the sound.

English	Attested	Computed
a	آ	ا
capitalism	كَابِتَالِيزْم	كَابِتَالِيزْم
capitalist	كَابِتَالِيسْت	كَابِتَالِيسْت
ideology	آيْدِيُولُوجِي	آيْدِيَالَجِي
individual	إِنْدِيْدِيُوَال	إِنْدِفَحُول
nature	تَيْشَر	تَيْشَر
psychoanalytic	سَايْكُوَأْتَالِيْتِيك	سَايْكُوَأْتَلِيك
structures	سْتَرْكُشُورْز	سْتَرْكُشَرْز
universal	يُونِيْزْسَل	يُونِيْفَرْسَل
utopianism	بُتُوِيَانِيْزْم	بُتُوِيِيْزْم

Table 5.1: Attested forms that differ from the computed forms in their letters.

5.5 The rules in order

The vowel rules are applied in a fixed order for each vowel phoneme, and the first rule that fires decides the mark and the vowel letters.

1. Schwa (AH) takes the colour of its spelling letter: *e*, *i* and *y* give kasra, *o* gives damma, *a* gives fatha, with fatha also for the vowel pairs *ia* and *io* and for word-final *-em*. A stressed AH spelled with *o* is written with damma and waw.
2. A rhotic vowel (ER) is written with fatha when unstressed, with the exception of the prefix *inter-*, which keeps kasra. Stressed ER takes its mark from the spelling letter, but *ea* gives fatha (*learn*).
3. The vowels AA and AO, when the spelling letter is *o*, are written with damma and waw (*on*, *Rollins*).
4. A stressed AE that is not word-initial gains an alef (*lack*).
5. A stressed EH or IH in an open syllable spelled with *e* or *i* gains a yeh (*Helen*, *cinema*).
6. An unstressed IY or UW in the first syllable is written short.
7. The vowel UH spelled with a written *oo* or *ou* keeps a waw (*book*, *good*, *foot*).

Each rule can be traced to specimen words, and each has exceptions among the attested forms that are listed in Table 5.1.

5.6 Double o and the word book

The last rule came from a correction by the author. A composed example wrote *book* as بُوْكَ, following the sound, and the author preferred بُوْكَ on the ground that English writes two vowel letters and the notation should show them. The author also noted that without the waw an Arabic reader might hear a different vowel. The same reasoning applies to *look*, *good*, *foot* and *wood*, and the toolkit now writes all of them with a waw. It does not yet extend to every doubled vowel, because the specimens contain no attested case of other doubled letters, apart from *ee*, which already receives a yeh as a long vowel.

5.7 Hosted vowels after a glide

When a vowel follows a glide, the mark is placed on the glide, so no separate hamza seat is needed. This accounts for *flower*, *power* and *hour*, where the diphthong in the first syllable ends in a waw glide and the unstressed syllable takes its fatha or kasra on that waw. The same device was already used after ي, as in *utopianism*.

5.8 Ng and the hard g

In the dictionary, *finger*, *anger* and *English* have an ng sound followed by a hard g. Written as separate sounds this gives a doubled ghain, which no specimen shows. The toolkit therefore writes the nasal and one ghain, as in اِنْغَلِيشْ. Words such as *singer* and *longer*, where the dictionary has only the nasal, are written the same way, which is a merger the notation accepts.

5.9 Examples

Table 5.2 gives words from the specimens and words that were not in them, written by the rules. The specimen words reproduce their attested forms. The others show how the rules extend.

English	Arabic script	Respelling
psychology	سَايْكُولُوجِي	<i>saykaalujee</i>
concept	كُونْسِبْتُ	<i>kaansept</i>
Rollins	رُولِنْز	<i>raalinz</i>
on	أُونْ	<i>aan</i>
cinema	سِينَمَا	<i>sinumu</i>
Helen	هِيلُنْ	<i>helun</i>
repression	رِبْرِشْنْ	<i>reepreshun</i>
social	سُوشِلْ	<i>sohshul</i>
nation	نَيْشْنْ	<i>neyshun</i>
system	سِيسْتَمْ	<i>sistum</i>
lack	لَاكْ	<i>lak</i>
part	بَارْتْ	<i>paart</i>
flower	فَلَاوَرْ	<i>flower</i>
hour	أَوَرْ	<i>ower</i>
power	بَاوَرْ	<i>power</i>
tower	تَاوَرْ	<i>tower</i>
finger	فِنْغَرْ	<i>fingger</i>
English	إِنْغِلِشْ	<i>ingglish</i>

Table 5.2: Words that exercise the spelling-conditioned rules.

Chapter 6

Architecture of the Toolkit

6.1 Modules

The toolkit is a Python package, `arabglish`, with one module per concern.

tables Sound-to-letter tables and the profile object. Every open decision is a parameter of `make_profile`.

lexicon Pronunciation lookup and alignment of vowel phonemes to vowel letters.

g2p A small spelling-to-sound fallback for words the dictionary lacks.

translit The forward transliterator, mark density, numbers and attested forms.

reader The reverse reader and round-trip measurement.

lint, corpus Inventory checking and specimen statistics.

guide The Latin respelling used in teaching glossaries.

cli The command line.

The command `arabglish` offers the subcommands `fwd`, `rev`, `lint`, `stats`, `glossary` and `roundtrip` (Appendix A). The original wrapper script, `main.py`, runs unchanged through a small compatibility module.

6.2 Pronunciation source

Pronunciations come from the CMU Pronouncing Dictionary through the `cmudict` package, which holds 126,052 entries of General American English. Lookup proceeds in order:

1. an override file, `pron_overrides.tsv`, for words the dictionary lacks or where a specimen needs a specific reading (*psychocinema*, *psychoanalytic*, *utopianism*, *disemvoweling*);
2. the dictionary, using its first pronunciation, and for the reverse index all pronunciations;
3. possessives built from the stem with the regular suffix;
4. the spelling fallback, which logs a warning so that weak spots are visible.

6.3 Replacing the pronunciation source

The notation is tied to one accent by this choice: General American, with its vowels and rhotics. A different source changes the Arabic output for many words. Pronunciation lookup is isolated in the single function `pronounce`, which returns a phoneme list and a source label, so another dictionary can be substituted without touching the rest. Candidates include the eSpeak NG phonemizer, which also covers unknown words and offers British voices, and the IPA transcriptions in Wiktionary. Which accent the book should follow is an open question for the author (Chapter 13).

6.4 Determinism

For a given option set the output is deterministic. Mark density is applied as a final filter over the fully marked form, so changing density never changes a letter: this is checked by a test over a sample text.

6.5 Module by module

tables.py The sound key as data: consonant letters, vowel marks and letters, fixed spellings of frequent words, punctuation, Arabic-Indic digits and the `Profile` object that records the open decisions.

g2p.py A small spelling-to-phoneme fallback. It handles the common English digraphs, the magic-e pattern and a few suffixes, and it assigns primary stress to the first vowel. It is deliberately simple, because every word that reaches it is logged and can be added to the override file.

lexicon.py The pronunciation layer. For a word it tries the override file, then the CMU dictionary, then a possessive rule, then the fallback, and it returns the phonemes with the name of the source. It also aligns the vowel letter groups of the spelling to the vowel phonemes.

translit.py The forward transliterator: segments, vowel placement, sukoon, density, attested forms, digits and text tokens.

reader.py The reverse reader and its index.

lint.py The inventory checker and the extended-letter repair.

corpus.py Statistics over specimens: letter counts, mark counts and inventory.

guide.py The Latin respelling used for teaching rows.

cli.py The command line.

6.6 Data files

Two small tab-separated files sit beside the code. `pron_overrides.tsv` lists words whose dictionary pronunciation is replaced, mainly the author's coinages and technical words (*psychocinema*, *disemvowel-ing*). `attested.tsv` records word forms taken from the specimens, with the source of each. Both are plain text, so they can be edited by hand and reviewed in a diff.

6.7 Pronunciation source

All pronunciation lookups go through a single function, `pronounce`. The dictionary is read once and cached. Replacing it with another source, such as a British dictionary, a machine-readable dictionary in a

different phone set, or a table of the author's own readings, means changing that function and a mapping from the new phone set to the ARPABET symbols used by the tables. Nothing downstream needs to change.

6.8 Diagnostics

The fallback logs every word it handles, and the unknown-phoneme path logs a warning, so weak spots are visible. The attested-form comparison is a script, not a test, so that it can be rerun and read. The test suite then fixes the numbers it produces, so regressions fail loudly.

6.9 Testing

The test suite has three kinds of test. Unit tests check single rules on single words, such as the waw of *book* and the silent possessive apostrophe in *Rollins'*. Invariant tests check properties that must always hold, such as that changing density never changes the unmarked skeleton. A ratchet test checks the agreement with the attested forms against fixed lower bounds, which are raised whenever the rules improve.

Chapter 7

The Forward Transliterator

7.1 From words to segments

For each word the engine obtains a phoneme list, aligns vowel phonemes to the vowel letter groups of the spelling when their counts agree, and builds a list of segments. Each segment is a letter with a kind (consonant, long vowel letter, glide, seat) and an optional mark.

Consonant phonemes append their letters, digraphs such as نُش appending two segments of one unit. Vowel phonemes attach their mark to the last consonant, or to a preceding ي, or to a new hamza seat, and then append their vowel letters. Rhotic vowels append ر.

7.2 The sukoon pass

After all segments are placed, every consonant or glide without a vowel receives sukoon. Long vowel letters and seats do not. The resulting form is the full-density spelling.

7.3 Mark density

A final filter produces the other densities, so density can never change a letter.

full all marks, including sukoon on every vowelless consonant.

medial sukoon removed from the end of the word.

vowels fatha, damma and kasra only.

none no marks. This is the consonantal skeleton with vowel letters, the form requested in the earliest experiments.

7.4 Attested forms

With `--attested` the engine prefers a form recorded in a specimen whenever the word occurs there. The list is built by aligning each specimen token to its English word and asserting that token counts match (`scripts/build_attested.py`). Page forms are listed before typed forms.

7.5 Demonstration

Table 7.1 shows one sentence of the page under several settings. The first row is the page itself.

Setting	Output
full marks	إِنْ ذِسْ بَارْتْ أَفْ هِيلِنْ رُولَنْزْ سَايْكُوسِيْنِمَا، شِي دِلْفَرْ إِنْتُو دَا تَيْتَشَرْ أَفْ لَآءْ.
computed, full	إِنْ ذِسْ بَارْتْ أَفْ هِيلِنْ رُولَنْزْ سَايْكُوسِيْنِمَا، شِي دِلْفَرْ إِنْتُو دَا تَيْتَشَرْ أَفْ لَآءْ.
medial sukoon	إِنْ ذِسْ بَارْتْ أَفْ هِيلِنْ رُولَنْزْ سَايْكُوسِيْنِمَا، شِي دِلْفَرْ إِنْتُو دَا تَيْتَشَرْ أَفْ لَآءْ.
vowel marks only	إِنْ ذِسْ بَارْتْ أَفْ هِيلِنْ رُولَنْزْ سَايْكُوسِيْنِمَا، شِي دِلْفَرْ إِنْتُو دَا تَيْتَشَرْ أَفْ لَآءْ.
unmarked	إِنْ ذِسْ بَارْتْ أَفْ هِيلِنْ رُولَنْزْ سَايْكُوسِيْنِمَا، شِي دِلْفَرْ إِنْتُو دَا تَيْتَشَرْ أَفْ لَآءْ.
v as feh, g as jeem	إِنْ ذِسْ بَارْتْ أَفْ هِيلِنْ رُولَنْزْ سَايْكُوسِيْنِمَا، شِي دِلْفَرْ إِنْتُو دَا تَيْتَشَرْ أَفْ لَآءْ.

Table 7.1: One sentence in several settings. The second row is the computed form with the defaults.

7.6 The density ladder

The four densities are best seen on a single sentence. Each line contains the same letters; only the marks differ.

Density	Result
full	دَا لَتَرْزْ تَشْيِيْنَجْ، بَتْ دَا وُرْدَرْ رَمِيْنْ.
medial	دَا لَتَرْزْ تَشْيِيْنَجْ، بَتْ دَا وُرْدَرْ رَمِيْنْ.
vowels	دَا لَتَرْزْ تَشْيِيْنَجْ، بَتْ دَا وُرْدَرْ رَمِيْنْ.
none	دَا لَتَرْزْ تَشْيِيْنَجْ، بَتْ دَا وُرْدَرْ رَمِيْنْ.

Table 7.2: One sentence at each density.

7.7 Worked passages

The following passages are public-domain texts and short originals, set with the default options. Each block shows the Arabic-script form, a Latin respelling that is only a reading aid, and the English. They illustrate how the rules behave on running prose and not on isolated words, and they include some of the weaknesses that Chapter 13 lists.

7.7.1 Children's prose

أَلِسْ وَازْ يَغْنِغْ تُو غَتْ فِيرِي تَابَرْدُ أَفْ سِتْنِغْ بَايْ هَرْ سِسْتَرْ أُونْ دَا بَانْغْكَ، أُنْدُ أَفْ
هَافْنِغْ تُو غَتْ دُو.

ألس واز بغنغ تو غت فيري تايرد أف ستنغ باي هر سستر أون ذا بانغك، أند أف
هافنغ نوغ تو دو.

*alus waaz bigining too get verree tayerd uv siting bay her sister aan dhu bangk und uv having
nothing too doo*

Alice was beginning to get very tired of sitting by her sister on the bank, and of having nothing
to do.

وُونْسْ أُوْرْ تَوَايْسْ شِيْ هَادْ بِيْتْ إِنْتُو دَا بُوْكَ هَرْ سِسْتَرْ وَازْ رِيْدِنِغْ، بَتْ إِيْثْ هَادْ تُو
يَكْتَشَرْزْ أُوْرْ كُونْفَرْسِيْشَنْزْ إِنْ إِيْثْ.

وونس أور توایس شی هاد بیت إنتو ذا بوك هر سستر واز ریدنغ، بت إیث هاد تو
بکتشرز أور کونفرسیشنز إِنْ إیث.

*wuns awr tways shee had peept intoo dhu buk her sister waaz reeding but it had noh pikcherz
awr kaanverseyshunz in it*

Once or twice she had peeped into the book her sister was reading, but it had no pictures or
conversations in it.

أُنْدُ وَتْ إِيْزْ دَا يُوْسْ أَفْ أْ بُوْكَ، تُوْتْ أَلِسْ، وَتَاوْتْ يَكْتَشَرْزْ أُوْرْ كُونْفَرْسِيْشَنْزْ؟
أُنْدُ وَتْ إِيْزْ دَا يُوْسْ أَفْ أْ بُوْكَ، تُوْتْ أَلِسْ، وَتَاوْتْ يَكْتَشَرْزْ أُوْرْ كُونْفَرْسِيْشَنْزْ؟

und wut iz dhu yoos uv u buk thawt alus withowt pikcherz awr kaanverseyshunz

And what is the use of a book, thought Alice, without pictures or conversations?

7.7.2 Scripture

إِنْ دَا يَغْنِغْ عُودْ كَرِيْتْدُ دَا هِيْفِنْ أُنْدُ دَا أَرْتْ.

in dhu bigining gaad kreeeytud dhu hevun und dhu erth

In the beginning God created the heaven and the earth.

أُنْدُ دَا أَرْتْ وَازْ وَتَاوْتْ فُوْرْمْ، أُنْدُ فُوْيدْ؛ أُنْدُ دَارْكِيسْ وَازْ أَبُوْنْ دَا فَيْسْ أَفْ دَا دِيْبْ.

und dhu erth waaz withowt fawrm und voyd und daarknus waaz upaan dhu feys uv dhu deep

And the earth was without form, and void; and darkness was upon the face of the deep.

أُنْدُ عُودْ سِيْدْ، لِيْثْ دِرْ يِيْ لَآيْثْ: أُنْدُ دِرْ وَازْ لَآيْثْ.

und gaad sed let dher bee layt und dher waaz layt

And God said, Let there be light: and there was light.

7.7.3 Verse

شَالْ آيْ كُمْپِرْ دِيْ تُوْ أْ سَمَرْزْ دِيْ؟

shal ay kumper dheer too u summerz dey

Shall I compare thee to a summer's day?

دَاوْ آرْت مُورْ لَفْلِي أَنْدْ مُورْ تَمُورْت.
dhow aart mawr luvlee und mawr temprut
 Thou art more lovely and more temperate.

رُوفْ وَنْدَزْ دُو شَيْكْ دَا دَارْلِيغْ بَدَزْ أَفْ مَي، أَنْدْ سَمَرْزْ لِيْسْ هَاثْ أَوْلْ تُو شُورْتْ أْ دَيْتْ.
ruf windz doo sheyk dhu darling budz uv mey und sumerz lees hath awl too shawrt u deyt
 Rough winds do shake the darling buds of May, and summer's lease hath all too short a date.

وُونْسْ أْبُونْ أْ مِدَتَايْتْ ذِرِيرِي، وَايْلْ آيْ بَانْدَرْدْ، وَيْلْ أَنْدْ وِيرِي،
wuns upaan u midnajt driree wayl ay paanderd week und wiree
 Once upon a midnight dreary, while I pondered, weak and weary,

أَوْقَرْ مِينِي أْ كَوَيْتْ أَنْدْ كِيرِيْسْ فُولْيُومْ أَفْ قَرْعُوتِيْنْ لُورْ.
ohver menee u kweynt und kyureeus vaalyoom uv fergaatun lawr
 Over many a quaint and curious volume of forgotten lore.

7.7.4 A long sentence

وِنْ إِنْ دَا كُورْسْ أَفْ هُيُومَنْ إِفْنَسْ، إِيْثْ يَكْمَرْ نِيْسِيْسِرِي فُورْ وُونْ بِيْلْ تُو دِرُولْفْ
 دَا بُلِيْتِكْلْ بَانْدَزْ وِشْ هَاْفْ كُنِكْتِيْدْ ذِمْ وَدْ أُوْدَرْ، أَنْدْ تُو أَسُومْ أُمُونْغْ دَا بَاوَرَزْ أَفْ دَا
 أَرْتْ، دَا سِيْبَرِيْتْ أَنْدْ إِيْكُولْ سَتِيْسَنْ تُو وِشْ دَا لُورْ أَفْ تِيْسَرْزْ أَنْدْ أَفْ تِيْسَرْزْ عُودْ
 إِيْتَايْتْلْ ذِمْ، أْ دِيْسِيْنْتْ رِشِيْكَتْ تُو دَا أُيْسِيْنَرْزْ أَفْ مَانْكَايْنْدْ رِكَوَايِرْزْ دَتْ دِيْ شُودْ دِكْلِرْ
 دَا كَارَزْ وِشْ إِيْمِلْ ذِمْ تُو دَا سِيْبَرِيْسَنْ.
wen in dhu kawrs uv hyoomun ivents it bikumz nesuseree fawr wun peepul too dizaalv dhu pulitukul bandz wich hav kunektid dhem widh unudher und too usoom umung dhu powerz uv dhu erth dhu sepereyt und eekwul steyshun too wich dhu lawz uv neycher und uv neycherz gaad entaytul dhem u deesunt rispekt too dhu upinyunz uv mankaynd reekwayerz dhat dhey shud dikler dhu kaazuz wich impel dhem too dhu sepereyshun

When in the Course of human events, it becomes necessary for one people to dissolve the political bands which have connected them with another, and to assume among the powers of the earth, the separate and equal station to which the Laws of Nature and of Nature's God entitle them, a decent respect to the opinions of mankind requires that they should declare the causes which impel them to the separation.

7.7.5 Narrative

دَا تُرَاْفَلَرْ كَيْمْ دَاوُنْ فُرُومْ دَا سِتَارْ سِيْرَسْ تُو آوَرْ لِيْتْلْ أَنْتْ هِلْ، أَنْدْ مِرَزْدْ دَا أَوْرْتْ وَدْ هِرْ فَنَعَزَرْ.
dhu travuler keym down frum dhu staar sirius too ower litul ant hil und mezherd dhu erth widh hiz fingerz
 The traveller came down from the star Sirius to our little ant hill, and measured the earth with his fingers.

هِي قَاوُنْدُ دَا يَيْتِلْ سَمُولْ، بَتْ هِي قَاوُنْدُ دِمْ كُيْرِيَسَنْ، اَنْدْ هِي اَسْكُتْ دِمْ مِني
 كُوسْتَسَنْزْ اَبَاوْتْ دِرْ لِيْفَرْ.

*hee fownd dhu peepul smawl but hee fownd dhem kyureeus und hee askt dhem menee kweschunz
 ubowt dher livz*

He found the people small, but he found them curious, and he asked them many questions
 about their lives.

7.8 Names, dates and numbers

Proper names are looked up like other words, and the fallback handles names the dictionary lacks. Numbers can be written in Arabic-Indic digits, in Latin digits or spoken, depending on the `digits` option. The spoken form is useful for text meant to be read aloud.

هِيلِنْ رُولِيْنزْ لِيْفَرْ اِنْ هَالِفَاكْسَنْ، اَنْدْ هِرْ بْرُوْدَرْ فِكْتَرْ لِيْفَرْ اِنْ فَنَكُوْفَرْ.

helun raalinz livz in halifaks und her brudher vikter livz in vankooover

Helen Rollins lives in Halifax, and her brother Victor lives in Vancouver.

تُوْرْدِيْ، دَا فِفْتْ اُفْ اُوْكُتُوْبَرْ، اَتْ سِيْفِيْنْ يَرْدِيْ، دَا تَرْيِنْ قُرُوْمْ تَرْوُتُوْ اَرَايْفَرْ.

toozdee dhu fifth uv aaktohaber at sevun therdee dhu treyn frum teraantoh erayvz

Tuesday, the fifth of October, at seven thirty, the train from Toronto arrives.

دِرْ اَرْ ٥٢ وُزْدَرْ اُونْ دِسْ يِيْجْ اَنْدْ ٣ اُفْ دِمْ اَرْ تَمْبَرَرْ.

dher aar werdz aan dhis peyj und uv dhem aar numberz

There are 25 words on this page and 3 of them are numbers.

اِنْ ٦٢٠٢ اِتْ كُوسْتْ ٠٥٢,١ دُولَرَرْ.

in it kaast daalerz

In 2026 it cost 1,250 dollars.

7.9 Reading the results

Three habits of the output are worth noting. Common words always take their fixed form, so *the*, *of*, *and* and *is* look the same everywhere. Stressed vowels are usually marked more heavily than unstressed ones, which gives longer words a visible rhythm. And function words with weak forms in speech, such as *to* and *a*, keep their fixed spellings and are not reduced.

Chapter 8

The Reverse Reader

8.1 Why the notation is many to one

Several English spellings share one sound, and several English sounds share one letter. The notation therefore cannot return a unique English word without help. *knight* and *night* are written identically, and so are *write* and *right*. Unstressed vowels that are optional marks add further collisions, as do the shared letters ب (for *b* and *p*, and for *v* under the *ḥ* setting) and ج or غ (for sounds merged with *j*).

8.2 Reader design

The reader indexes every dictionary word by its Arabic skeleton, the letters without marks, and by a coarser key without vowel letters. For an input word it returns candidates in tiers: attested forms first, then skeleton matches ranked by corpus frequency with a preference for those whose marks agree with the input, then matches on the coarse key. It reports a ranked list and never a single guess.

Table 8.1 shows pairs that the notation cannot separate. The original word is among the candidates but not always first.

English	Spelling	Top candidates	Rank of original
knight	تَايَيْتْ	night, knight, nite	2
planet	بَلَانَيْتْ	plant, planet, plante, pylant	2
their	ذُرْ	their, there, thur	1
write	رَايَيْتْ	right, write, wright, riot	2

Table 8.1: Words whose spelling is shared with another English word.

8.3 Measured ambiguity

Two measures describe the notation, both over the 5000 most frequent English words that the dictionary covers.

- **Collisions.** 9.8% of these words share their fully marked form with at least one other word, and 14.2% share their unmarked form. Table 8.2 shows the largest classes.
- **Reading back.** For the 2000 most frequent words, the reader returns the original as its first candidate 92% of the time and among its first five 99%. The mean candidate list has 8.8 entries. This

figure is optimistic by construction: the test words are frequent, frequency ranks the candidates, and the reader uses the same dictionary as the writer.

Unmarked form	Words sharing it
بت	but, put, bit, bet, pet, butt, pit
بول	ball, paul, pool, bowl, poll, pole
بي	be, pay, b, p, bay
بن	been, ben, pen, pin, bin
سي	see, say, c, sea, se
أن	an, un, ann, anne

Table 8.2: Largest classes of words that share an unmarked spelling.

8.4 Consequence

Exact recovery of English spelling from the Arabic form is not available. A phonetic reading is always available. A lossless round trip would require extra information: surrounding context, a language model over the candidates, or optional disambiguation marks. The tools therefore separate recovering a pronunciation from recovering a spelling, and they report ambiguity in place of hiding it.

8.5 A reading in detail

The reader treats each token separately. The example below passes the earlier sentence back through it, with the specimen forms switched off. Fixed spellings of frequent words are always available, because they are part of the notation.

Written	Top candidates	Reader tier
دَا	the, with, they, though	attested
لِتَرْزُ	letters, walters	skeleton
تَسْبِيحٌ،	change	skeleton
بِتْ	but, butt, put, bit	skeleton
دَا	the, with, they, though	attested
وُزْدُرْ	words, awards, roads, yards	skeleton
رِمَيْنْ.	remain, roman, romania, romney	skeleton

Table 8.3: Reader output for one sentence: the first four candidates for each token.

In most rows the intended word is first. The token *إِزْ* is shared by *is* and *as*, and the reader cannot choose between them without context. Such pairs are the main cost of the compact notation.

8.6 Using context

The reader treats words independently. Choosing among candidates with a language model over the whole sentence is the natural next step, because each candidate list is short and most ambiguities resolve easily from context. The data structures already return ranked lists, so a context model can be added as a reranking step without changing the index.

8.7 Tiers and their meaning

The tier attached to each candidate says how closely it matches. An attested or fixed form is an exact recovery. A skeleton match means the letters agree and the marks are compatible. A consonant match means only the consonant skeleton agrees, and it is offered when the other tiers have little to say. A reader that shows the tier lets the user judge how much weight a candidate deserves.

Chapter 9

The Linter

9.1 Purpose

The specimens drift. The Atlas text uses Persian letters, the Micromegas text uses one of them, a web export replaced some letters with the Unicode replacement character, and typed passages occasionally place two vowel marks on one letter. A notation that depends on a small inventory needs a mechanical check that the inventory is respected. The linter in `arabglish.lint` provides that check.

9.2 Checks

The linter reports each problem with its position, the offending character, a code and a message. The codes are as follows.

extended-letter A letter outside the standard Arabic block, such as پ, چ, ژ, گ or ف. A suggested replacement is attached.

corrupt The replacement character U+FFFD. The original letter is lost and cannot be repaired automatically.

orphan-mark A vowel mark or sukoon with no base letter.

double-vowel Two vowel marks on one letter.

sukoon-and-vowel A sukoon and a vowel mark on one letter.

shadda A shadda. The system does not mark English doubling, so a shadda is an error.

tatweel The elongation character, which is not part of the system.

unknown-arabic Any other Arabic-block character outside the inventory.

Punctuation, Latin text, digits and the Arabic comma, semicolon and question mark are accepted.

9.3 Repair

Only the deterministic part of the repair is automated. The function `fix` rewrites extended letters to the core inventory: پ becomes ب, ف becomes ف, گ becomes غ, چ becomes چ, ژ becomes ز, and the Persian forms of kaf and yeh become the Arabic ones. This follows the decisions in Chapter 3.

If a different decision is adopted for *v* or *g*, the replacement table is the one place to change. Corrupt characters, orphan marks and double marks are reported but left for the author, because several repairs are possible.

9.4 Results on the specimens

Table 2.1 lists the issue counts. The page transcript and typed passage are clean. The Atlas text has 125 issues, nearly all extended letters. Running `fix` on it produces text that passes the linter, which is a convenient way to bring early material into the current inventory.

9.5 A worked example

The line below is a deliberately damaged piece of text. It contains four extended letters, a stray shadda and an orphan mark. The linter lists every problem with its position, and `fix` repairs the part that has a single correct repair.

Character	Code	Suggested repair
گ	extended-letter	غ
ک	extended-letter	ك
ف	extended-letter	ف
چ	extended-letter	تش
پ	extended-letter	ب
ّ	shadda	–
ّ	orphan-mark	–

Table 9.1: Linter output for a damaged line.

Before and after the deterministic repair:

دَا غُود بُوكْ إِزْ قَرِي نَشِيْبْ ّ → دَا گُود بُوكْ إِزْ قَرِي چِيْپْ ّ

The shadda is not repaired because there is no single correct reading of it. It may be an error or a deliberate marking of a doubled consonant, and the author decides which.

9.6 Using the linter in a workflow

The linter returns a nonzero status when issues are found, so it can run in a script before typesetting or before a text is added to the specimen set. A typical sequence is to repair the extended letters automatically, rerun the linter, and read the remaining items by hand. Because the linter reports positions, an editor can jump to each item.

9.7 What the linter does not check

The linter checks the inventory and the order of marks. It does not check that the spelling is a good rendering of any English word, because the notation has no single correct spelling for many words. Checking against a pronunciation is the job of the forward transliterator, and the evaluation in Chapter 10 compares its output with attested forms.

Chapter 10

Evaluation

10.1 What can be measured

The only ground truth available is the set of 45 word forms written by the author in the typed passage and the handwritten page. The computed form of each word is compared with the attested form at two levels: exact agreement of letters and marks, and agreement of letters once marks are removed.

10.2 Results

Of 45 attested words, 33 (73%) are reproduced exactly and 35 (78%) agree at the level of letters. Ten words differ. Table 5.1 lists them.

The differences fall into five groups. A lexical alef appears in forms such as *capitalism*, where the author writes a long vowel the dictionary reading does not require. The choice for *v* differs in *individual* and *universal*, where the typed passage uses ب and the default is ف. Some alignments of spelling to sound fail on longer words such as *individual* and *ideology*. The seat letter for an initial vowel differs in *a*. Finally, the digraph *ch* in *nature* is written نش on the page and ش in the typed passage, so one of the two attested forms must disagree with any single rule.

10.3 Interpretation

The rules of Chapter 5 were fitted to these same specimens. The agreement is therefore a description of how well the rules summarise the evidence and is not a validation on unseen text. Held-out specimens are the main need; Chapter 13 returns to this.

10.4 Regression ratchet

The test suite fixes lower bounds on the two agreement figures (0.77 for letters and 0.72 for exact forms). A change to the rules that lowers either figure fails the tests. A change that raises them is accepted and the bound can be moved up. This keeps later work from silently undoing earlier fits.

10.5 Evaluating the reader

The reader is evaluated separately in Chapter 8. It is measured with the attested tier switched off, so that specimen words do not recognise themselves, and on frequent words only.

10.6 Sensitivity to the open switches

The two main open switches change few attested words, so the agreement figures barely move with them. Table 10.1 shows all four combinations of the settings for *v* and hard *g*.

<i>v</i>	hard <i>g</i>	Exact	Letters
ف	غ	33	35
ف	ج	33	35
ب	غ	33	35
ب	ج	33	35

Table 10.1: Agreement with the 45 attested forms under each setting of *v* and hard *g*.

The equal numbers do not mean the settings are equivalent. The attested set contains only two *v* words with disagreeing letters, and no word with a hard *g*, so the specimens cannot rank the options. This is the reason both stay as switches.

10.7 Agreement on newly composed examples

After the main rules were fixed, three further sentences were composed in the notation and reviewed by the author, who accepted them with one correction (*book*). They were then compared with the toolkit word by word. Of 25 words, 22 are reproduced exactly. Table 10.2 lists the others.

English	Approved	Computed
words	وَرْدَر	وُورْدَر
recognise	رِكْعَنَائِر	رِيكْعُنَائِر
familiar	فَمَلِير	فَمِيلِير

Table 10.2: Words in the approved sentences that the toolkit writes differently.

These sentences were composed after the rules and not before, but they were composed by the same system of reasoning and are not independent evidence. Their value is as a check that the rules produce forms the author finds acceptable on unseen words. One difference, *words*, concerns the vowel before a cluster, and two more concern the open-syllable yeh in *recognise* and *familiar*. Each is a candidate for a rule change once the author has decided which form is wanted.

10.8 Error analysis

The remaining disagreements with the attested forms are of three kinds. Some are lexical, such as the alef in *capitalism*, and no rule predicts them. Some come from alignment, where the number of vowel letter groups in the spelling does not match the number of vowel sounds. And some are choices that the specimens contradict among themselves, such as *nature*. Only the second kind can be reduced by better software, and the first and third require decisions or more data.

10.9 A proper held-out test

A real test needs words the rules have not seen. The simplest design is to give the author a list of unseen words, have each written by hand in the notation without the tool, and compare. Fifty words chosen across frequency bands, with a few names and a few long technical words, would be enough to show which rules generalise and which were overfitted. The scripts already accept such a file and print the same tables as for the present specimens.

Chapter 11

Specimen Glossary

This chapter applies the forward transliterator to the opening of the typed passage. Each entry has three lines: the Arabic-script form with the default settings, a Latin respelling generated from the Arabic form by the guide tool, and the English original. The respelling is a reading aid for the author and for readers who do not read Arabic script. It is not part of the notation.

دَا رِبْرِشَنُ أَفْ لَآكْ أَأَنْدُ إِتْسَنُ كَلْشَرْلُ سِيْمُتْمَزْ
dhu reepreshun uv lak und its kulcherul simptumz
The repression of lack and its cultural symptoms.

إِنْ دِسْ بَارْتْ أَفْ هِيلِنْ رُولِنَرْ سَايْكُوسِيْنِمَا
in dhis paart uv helun raalinz saykohsinumu
In this part of Helen Rollins' psychocinema,

شِي دِلْفَرْ إِتْثُو دَا تَيْشَرْ أَفْ لَآكْ
shee delvz intoo dhu neycher uv lak
she delves into the nature of lack.

11.1 Ambiguity and context

Some distinctions disappear in the notation. *Is* and *as* share a written form, and so do *night* and *knight*. Taken one at a time, such collisions show where the system's limits lie, and Chapter 8 measures them. Their practical weight is a separate question, because readers rarely meet words in isolation. In a sentence, the intended word is usually clear from its neighbours, as it is for homophones in spoken English, and a collision matters only where it also causes confusion in continuous prose. The lists below are therefore best read as evidence about the notation's resolution, and not as a prediction that each pair will cause trouble in use.

11.2 The reference vocabulary

The tables below group words by the feature they illustrate. Every form is generated by the toolkit with the default options, and every table is regenerated whenever the rules change, so the book cannot drift from the code.

11.2.1 Doubled o and related spellings

English	Arabic script	Respelling
book	بُوكْ	<i>buk</i>
look	لُوكْ	<i>luk</i>
good	عُودْ	<i>gud</i>
foot	فُوتْ	<i>fut</i>
wood	وُودْ	<i>wud</i>
food	فُودْ	<i>food</i>
moon	مُونْ	<i>moon</i>
school	سَكُولْ	<i>skool</i>
too	تُو	<i>too</i>
two	تُو	<i>too</i>
blue	بَلُو	<i>bloo</i>
true	تُرُو	<i>troo</i>

11.2.2 ee and ea

English	Arabic script	Respelling
see	سِي	<i>see</i>
tree	تَرِي	<i>tree</i>
green	غَرِينْ	<i>green</i>
sleep	سَلِيبْ	<i>sleep</i>
head	هَدْ	<i>hed</i>
bread	بَرْدْ	<i>bred</i>
read	رَدْ	<i>red</i>
meet	مِيتْ	<i>meet</i>
meat	مِيتْ	<i>meet</i>
dream	دَرِيمْ	<i>dreem</i>
dead	دَدْ	<i>ded</i>

11.2.3 Glides and diphthongs

English	Arabic script	Respelling
rain	رَيْنَ	<i>reyn</i>
day	دَيَّ	<i>dey</i>
light	لَايْت	<i>layt</i>
night	نَايْت	<i>nayt</i>
voice	فُؤَيْسَ	<i>voys</i>
flower	فَلَاوَر	<i>flower</i>
hour	أَوَر	<i>ower</i>
boy	بُؤَيَّ	<i>boy</i>
toy	تُؤَيَّ	<i>toy</i>
noise	نُؤَيْر	<i>noyz</i>
now	نَاو	<i>now</i>
how	هَاو	<i>how</i>

11.2.4 Short vowels

English	Arabic script	Respelling
cat	كَاتْ	<i>kat</i>
bed	بَدْ	<i>bed</i>
sit	سَيْتْ	<i>sit</i>
hot	هُوْتْ	<i>haat</i>
cut	كَتْ	<i>kut</i>
put	بُتْ	<i>put</i>
pen	پَنْ	<i>pen</i>
map	مَابْ	<i>map</i>
top	تُوبْ	<i>taap</i>
sun	سَنْ	<i>sun</i>

11.2.5 Silent e

English	Arabic script	Respelling
make	مَيْكَ	<i>meyk</i>
time	تَايْمْ	<i>taym</i>
home	هُومْ	<i>hohm</i>
use	يُوسْ	<i>yoos</i>
cake	كَيْكَ	<i>keyk</i>
like	لَايْكَ	<i>layk</i>
phone	فُؤَنْ	<i>fohn</i>
tune	تُونْ	<i>toon</i>

11.2.6 R-coloured vowels

English	Arabic script	Respelling
car	كَارَ	<i>kaar</i>
far	فَارَ	<i>faar</i>
her	هَرَ	<i>her</i>
bird	بِرْدَ	<i>berd</i>
turn	تَرْنَ	<i>tern</i>
word	وَرْدَ	<i>werd</i>
door	دَوْرَ	<i>dawr</i>
more	مَوْرَ	<i>mawr</i>
fire	فَايَرَ	<i>fayer</i>
tire	تَايَرَ	<i>tayer</i>

11.2.7 Consonants

English	Arabic script	Respelling
thing	ثِيْنَعْ	<i>thing</i>
think	ثِيْنَعْلُكْ	<i>think</i>
this	ذِسْ	<i>dhis</i>
that	ذَتْ	<i>dhat</i>
church	تَشْرُوشْ	<i>cherch</i>
change	تَشْيِيْنَجْ	<i>cheynj</i>
judge	جُجْ	<i>juj</i>
measure	مِيَزَرْ	<i>mezher</i>
vision	فِيْزَنْ	<i>vizhun</i>
singer	سِيْنَعَرْ	<i>singer</i>
finger	فِيْنَعَرْ	<i>fingger</i>
garden	عَاْرِدَنْ	<i>gaardun</i>

11.2.8 Clusters

English	Arabic script	Respelling
string	سْتِرِيْنَعْ	<i>string</i>
spring	سْتِرِيْنَعْ	<i>spring</i>
splash	سَبْلَاشْ	<i>splash</i>
street	سْتِرِيْتْ	<i>street</i>
strength	سْتِرِيْنَعْكُتْ	<i>strengkth</i>
twelfth	تَوَلْفُتْ	<i>twelfth</i>
asked	أَسْكُتْ	<i>askt</i>
texts	تِيْكُسْتَسْ	<i>teksts</i>

11.3 Teaching sequence

The materials in this chapter suit a staged course in which the supporting lines are withdrawn as the reader gains fluency.

1. **Paired reading.** Begin with the Arabic-script English beside the familiar English word, so that each new spelling is tied to a word the reader already knows.
2. **A small recurring vocabulary.** Practise a few dozen frequent words (*the, of, and, is, in, to*) until their written forms are recognised at once. Their fixed spellings make this easy.
3. **Short passages with the English covered.** Read short passages with the English line concealed, uncovering it only to check.
4. **Withdraw the respelling.** Drop the Latin respelling next, since it is an approximate aid. It introduces spelling conventions of its own (the *ee* of *reeding*, the *u* of *uv*), and a reader who leans on it too long learns those conventions instead of the notation.
5. **Lighter marks.** Finally, move from full marks to medial, vowel-only and unmarked text, using the density settings.

11.4 Reading practice

The following passages start with the simplest sentences and become longer. Readers learning the notation can cover the English and the Latin respelling and read the Arabic-script line aloud.

دَا بُوكُ إِزْ رِتِنْ إِنْ إِنْغِلِشْ.
 ذا بوك إز رتن إن إنغليش.
dhu buk iz ritun in ingglish
 The book is written in English.

دَا لَتَرَزْ تُشَيِّنْجْ، بَتْ دَا وَرْدَزْ رَمَيْنْ.
 ذا لترز تشينج، بت ذا وردز رمين.
dhu leterz cheynj but dhu werdz rimeyn
 The letters change, but the words remain.

أَ رِيدَرْ لَرَنْزْ دَا سَكْرِبَتْ أَندْ يَغْنَزْ تُو رِيكُغَنَائِزْ أَ فَمِيلِيَرْ فُويسْ.
 أ ريدر لرنز ذا سكربت أند بغنز تو ريكغنايز أ فميليير فويس.
u reeder lernz dhu skript und biginz too rekugnayz u fumilyer voys
 A reader learns the script and begins to recognise a familiar voice.

أَلِسْ وَازْ يَغْنِغْ تُو غَتْ فِيرِي تَايَرْدُ أَفْ سِتِنْجْ بَايْ هَرْ سِسْتَرْ أُونْ دَا بَانْغْكَ، أَندْ أَفْ
 هَافِنْجْ نُوتِنْجْ تُو دُو.
 ألس واز بغنغ تو غت فيري تايرد أف ستنج باي هر سستر أون ذا بانغك، أند أف
 هافنج نوthing تو دو.
alus waaz bigining too get verree tayerd uv siting bay her sister aan dhu bangk und uv having nothing too doo
 Alice was beginning to get very tired of sitting by her sister on the bank, and of having nothing to do.

وُونَسْ اُور تَوَايسْ شِي هَادْ بِيْتْ اِنتُو دَا بُوْكْ هِرْ سِسْتَرْ وَاَزْ رِيْدِيْعْ، بَتْ اِثْ هَادْ نُو
يَكْتَشَرَزْ اُورْ كُونْفَرَسِيْشَنَزْ اِنْ اِثْ.

وونس اور توایس شی هاد بیت ایتو دا بوک هر سستر واز ریدیع، بت ایت هاد نو
بکتشرز اور کونفرسیشنز اِن اِثْ.

*wuns awr tways shee had peept intoo dhu buk her sister waaz reeding but it had noh pikcherz
awr kaanverseyshunz in it*

Once or twice she had peeped into the book her sister was reading, but it had no pictures or
conversations in it.

اَنْدْ وِتْ اِزْ دَا يُوْسْ اَفْ اَبُوْكْ، تُوتْ اَلِسْ، وَتَاوْتْ يَكْتَشَرَزْ اُورْ كُونْفَرَسِيْشَنَزْ؟
اَنْدْ وِتْ اِزْ دَا يُوْسْ اَفْ اَبُوْكْ، تُوتْ اَلِسْ، وَتَاوْتْ يَكْتَشَرَزْ اُورْ كُونْفَرَسِيْشَنَزْ؟

und wut iz dhu yoos uf u buk thawt alus withowt pikcherz awr kaanverseyshunz

And what is the use of a book, thought Alice, without pictures or conversations?

اِنْ دَا يَغِيْنْعْ عُوْدْ كَرِيْتْدْ دَا هِيْفِنْ اَنْدْ دَا اَرْتْ.

in dhu bining gaad kreeeytud dhu hevun und dhu erth

In the beginning God created the heaven and the earth.

اَنْدْ دَا اَرْتْ وَاَزْ وَتَاوْتْ فُوْرْمْ، اَنْدْ فُوِيْدْ؛ اَنْدْ دَا رَكْنِسْ وَاَزْ اَبُوْنْ دَا قَيْسْ اَفْ دَا دِيْبْ.

und dhu erth waaz withowt fawrm und voyd und daarknus waaz upaan dhu feys uv dhu deep

And the earth was without form, and void; and darkness was upon the face of the deep.

اَنْدْ عُوْدْ سِيْدْ، لِيْثْ دِرْ يِيْ لَابِيْثْ: اَنْدْ دِرْ وَاَزْ لَابِيْثْ.

und gaad sed let dher bee layt und dher waaz layt

And God said, Let there be light: and there was light.

دَا تُرَاْفِلَرْ كَيْمْ دَاوُنْ فُرُوْمْ دَا سَتَارْ سِيْرْسْ نُو اَوْرْ لِيْثْ اَنْتْ هِلْ، اَنْدْ مِرَزْدْ دَا اَرْتْ وَدْ
هِرْ فَنْعَرَزْ.

*dhu travuler keym down frum dhu staar sirius too ower litul ant hil und mezherd dhu erth widh
hiz fingerz*

The traveller came down from the star Sirius to our little ant hill, and measured the earth with
his fingers.

هِيْ قَاوُنْدْ دَا يِيْبَلْ سُمُوْلْ، بَتْ هِيْ قَاوُنْدْ دِمْ كِيْرِيْسْ، اَنْدْ هِيْ اَسْكْتْ دِمْ مِيْنِيْ
كُوْسْتَشَنَزْ اَبَاوْتْ دِرْ لِيْفَرْ.

*hee fownd dhu peepul smawl but hee fownd dhem kyureeus und hee askt dhem menea kweschunz
ubowt dher livz*

He found the people small, but he found them curious, and he asked them many questions
about their lives.

Chapter 12

Typesetting and Input

12.1 Typesetting this book

The book is set with XeLaTeX. The Arabic-script specimens are short runs inside English text, so the bidi-rectional machinery of a full Arabic setup is not needed. The `\ar` macro starts a right-to-left run with the TeX_{XX}E_T primitives (`\beginR` and `\endR`) and selects DejaVu Sans, which carries the Arabic block and positions fatha, damma, kasra and sukoon correctly. Packages that usually do this work, `polyglossia` and `bidif`, were not available in the build environment, and LuaLaTeX failed on font lookup. The TeX_{XX}E_T route is small and has been checked by rendering pages and inspecting them.

Two cautions apply. A right-to-left run must not be broken across lines by hand, since the order of the pieces then depends on the line breaker. Letters and marks that arrive in the wrong order from a generator will render wrongly but silently, which is a reason to lint the text before typesetting it.

12.2 Input

The AutoHotkey script `toarabic.ahk` (Appendix C) is an input method: it maps one Latin key to one Arabic letter. That is useful for typing the notation by hand and unrelated to the sound key. A keyboard that matches the sound key would instead offer the consonant letters under the Latin keys that look most like the sounds (b and p on ب, f on ف, g on غ) and the three vowel marks and sukoon on spare keys. Such a layout could be generated from the tables in `tables.py`. It is listed with the open work.

12.3 Mark density in print

Fully marked text is long and visually dense. The `medial` setting removes sukoon at the ends of words, `vowels` keeps vowel marks only, and `none` removes all marks. Because density is applied as a last filter, the same source text can be set at several densities without changing any letters. Table 7.1 shows the settings side by side.

12.4 Fonts

The notation needs a font that carries the basic Arabic block and positions the three vowel marks and sukoon cleanly above and below their base letters. DejaVu Sans does this reliably in XeLaTeX. Other fonts with full Arabic support, such as Amiri or Noto Naskh, can be substituted, and it is worth checking

each by rendering a page, because some place the marks too close to the letters when several letters carry marks in a row.

12.5 Copying and pasting

Marks are separate code points that follow their base letter. Text copied from a web page or a word processor can lose or reorder them, and the linter's orphan-mark check catches the usual damage. Before a text is added to the specimens, it is worth running the linter once on the pasted result.

12.6 Printing the book

Because Arabic-script runs are short and set inline, the line breaker treats them as ordinary text, and long runs are better placed in their own paragraph or in a quotation block, as the worked passages in this book are. A run that must break across lines should be split between words, since splitting inside a word separates letters from their marks.

12.7 Keyboard layout

The AutoHotkey layout in the author's repository maps one Latin key to one Arabic letter, which is an input method. A layout designed from the sound key would put the letters under the Latin keys that look most like the sounds and place the marks on spare keys. A table-driven generator for such a layout would take its data from `tables.py`, so that a change of defaults changes the layout too. It is listed under open work, because it depends on the author's decision about how many marks should be typed by hand.

12.8 Accessibility

Screen readers read Arabic script with Arabic pronunciation rules, so notation text will be read as if it were Arabic. For an audience that needs speech, the Latin respelling or the English text should be offered next to it.

Chapter 13

Open Work

13.1 Pronunciation source

The forward transliterator reads pronunciations from the CMU Pronouncing Dictionary, a General American dictionary with about 126 000 entries, and falls back to a small spelling-based guesser for words it lacks. An earlier version of the system used a different pronunciation source, described as a kind of radio English, and the dictionary behind that version has not been identified. Which accent the notation should encode is an author decision. Candidates include another machine-readable dictionary (for example a British one with received pronunciation), or a table of per-word overrides for words where the author's own reading differs. The toolkit isolates the pronunciation source in one function, so changing it does not touch the sound key.

13.2 The decisions still open

Table 3.3 lists the switches. The most consequential is the hard *g* (غ or ج); *v* is now ف by the author's decision, with و available. The hard *g* can be settled by a statement from the author or by a larger sample of attested forms.

13.3 More specimens

All fitted rules come from one typed passage and one handwritten page. Held-out hand-written pages, written without reference to the tool, would allow a real test of the rules in Chapter 5.

13.4 Disambiguation

Chapter 8 showed that the notation merges some words. Optional disambiguation marks, or a language model that ranks reader candidates in context, would reduce this. Whether the notation should carry such marks at all is a design choice that the author has not yet made.

13.5 Further questions

Whether English doubling should ever be marked with shadda, how to write stress, how to treat names from other languages, and whether the system should have dialect profiles are all left open. None is needed for the present tools.

13.6 Priorities

The open questions differ in cost and in value. The list below is in a suggested order.

1. A held-out specimen: fifty unseen words written by hand without the tool. This tests every rule and costs an hour.
2. A decision on the stressed vowel of *words* (وَرْدٌ against the computed وَرْدٌ). For *written* the author chose the computed رِتْن over the first composed form رَتْن, which settles the unstressed vowel before a final *n* as kasra.
3. A decision on the open-syllable yeh in words such as *familiar* and *recognise*, where the stressed syllable is closed in speech but open in spelling.
4. A decision on whether to write the stress of English words.
5. A choice of pronunciation source, including regional dictionaries.
6. A context model for the reverse reader.
7. A keyboard layout generated from the tables.

13.7 Homographs

Words such as *read*, *lead* and *live* have more than one pronunciation. The forward transliterator takes the first dictionary entry. A part-of-speech tagger or a word-sense list could choose among them, and the author could instead mark the intended reading by hand. For now, a per-word override handles the cases the author cares about.

13.8 Proper names and coinages

Names the dictionary does not contain are handled by the fallback, whose weak spots appear whenever it logs a word. The override file is the place to record the author's reading of a name. A list of the first hundred such words from the author's own writing would improve the notation on the texts the author actually produces.

13.9 Dialects and accents

The dictionary reflects General American speech. The notation could carry an accent profile, as it already carries profiles for *v* and hard *g*, and a profile for a British, Canadian or other accent would change the phoneme source and perhaps a few rules. Nothing in the present design prevents this, but no specimen distinguishes them.

13.10 Concluding reflection

The project has a theoretical side that this book has kept apart from its tools. The author connects it to Deacon's account of symbolic reference and to the reading of film in the *Psychocinema* discussion, and neither is needed to use or evaluate the notation. The strongest version of the connection rests on the reader's experience. Reading English in an unfamiliar script makes the normally automatic work of recognition temporarily conspicuous. The reader sees that a word is being identified from a pattern of marks, that the pattern is a convention, and that the convention can be learned and then forgotten again as it becomes fluent. That observation stands on its own and gives the project a philosophical interest. Nothing in the readability of the notation depends on accepting a larger theory of language or of cinema.

Appendix A

Command Line Reference

The package is run as `python -m arabglish` with the source directory on the path. The original wrapper `main.py` is kept for compatibility and takes a text or an input file.

fwd English to Arabic script. Options: `--density` (full, medial, vowels, none), `--v` (b, f, context), `--g` (ghain, jeem), `--persian`, `--digits` (indic, latin, spoken), `--attested`, `--the-before-vowel`, `--text`, `-i`, `-o`.

rev Arabic script to ranked English candidates.

lint Check a text against the inventory; `--fix` rewrites extended letters.

stats Letter, mark and issue counts for a file.

glossary Three-line entries (Arabic, respelling, English).

roundtrip Measure reader recovery over frequent words.

Examples:

```
python3 main.py "the nature of lack"
python3 -m arabglish fwd --density none --text "the nature of lack"
python3 -m arabglish lint specimens/atlas.txt
```

The test suite runs with `pytest`. Tables for this book are regenerated by `scripts/make_tex_tables.py`.

A.1 Sample sessions

Forward transliteration with different densities:

```
$ python3 main.py "The book is written in English."
هـلـغـنـا نـا نـتـر زـا كـوب اـد
```

```
$ python3 -m arabglish fwd --density none \
    --text "The book is written in English."
هـلـغـنـا نـا نـتـر زـا كـوب اـد
```

Switching the open decisions:

```
$ python3 -m arabglish fwd --v b --text "very live"  
$ python3 -m arabglish fwd --g jeem --text "good game"
```

Checking and repairing text:

```
$ python3 -m arabglish lint specimens/atlas.txt  
$ python3 -m arabglish lint --fix specimens/atlas.txt -o atlas_fixed.txt
```

Rebuilding the book's tables and passages:

```
$ PYTHONPATH=src python3 scripts/make_tex_tables.py  
$ PYTHONPATH=src python3 scripts/make_passages.py  
$ cd monograph && xelatex main.tex && xelatex main.tex
```

Appendix B

Attested Forms

Forms written by the author in the specimens, with the form computed at the default settings. Source P is the handwritten page, T the typed passage.

English	Attested	Source	Computed	Match
a	أَ	T	أَ	differs
and	أَنْدَ	TP	أَنْدَ	exact
as	أَرْ	T	أَرْ	exact
both	بُوْثَ	T	بُوْثَ	exact
built	بِلْثَ	T	بِلْثَ	exact
capitalism	كَابِتَالِيْزْمَ	T	كَابِتَالِيْزْمَ	differs
capitalist	كَابِتَالِيْسْتَ	TP	كَابِتَالِيْسْتَ	differs
concept	كُوْنِسِيْبْتَ	T	كُوْنِسِيْبْتَ	exact
cultural	كُلْتَشَرْلَ	TP	كُلْتَشَرْلَ	exact
delves	دَلْفَرْ	TP	دَلْفَرْ	exact
denial	دِتَايَلَ	T	دِتَايَلَ	exact
explores	إِكْسْبُلُوْرَرْ	T	إِكْسْبُلُوْرَرْ	exact
for	فُوْرَ	T	فُوْرَ	exact
frames	فُرِيْمَرْ	T	فُرِيْمَرْ	exact
helen	هِيْلَنَ	TP	هِيْلَنَ	exact
ideology	آِيْدِيُولُوْجِيْ	T	آِيْدِيَالَجِيْ	differs
in	إِنَ	TP	إِنَ	exact
individual	إِنْدِيْوِيْوَالَ	T	إِنْدِيْوَالَ	differs
intersects	إِنْتَرِيْكِسْكَسَ	T	إِنْتَرِيْكِسْكَسَ	exact
into	إِنْتُوْ	TP	إِنْتُوْ	exact
it	إِتَ	T	إِتَ	exact
its	إِنْسَ	TP	إِنْسَ	exact
lack	لَاكَ	TP	لَاكَ	exact
nature	تِيْنَشَرْ	P	تِيْنَشَرْ	letters
nature	تِيْنَشَرْ	T	تِيْنَشَرْ	differs
of	أَفَ	TP	أَفَ	exact
on	أُوْنَ	T	أُوْنَ	exact
part	بَارْتَ	TP	بَارْتَ	exact
psychoanalytic	سَايْكُوْأَالِيْتِيْكَ	T	سَايْكُوْأَالِيْتِيْكَ	differs
psychocinema	سَايْكُوْسِيْنِيْمَا	TP	سَايْكُوْسِيْنِيْمَا	exact

English	Attested	Source	Computed	Match
psychology	سَايْكُولُوجِي	T	سَايْكُولُوجِي	exact
ramification	رَامِفِكَيْشَن	T	رَامِفِكَيْشَن	exact
repression	رِبْرِشَن	TP	رِبْرِشَن	exact
rollins	رُولِنَر	TP	رُولِنَر	exact
she	شِي	TP	شِي	exact
social	سُوشَل	T	سُوشَل	exact
structures	سُتْرُكُتُشَرُز	T	سُتْرُكُتُشَرُز	differs
symptoms	سِمِطَمَر	TP	سِمِطَمَر	exact
system	سِيسْتَم	T	سِيسْتَم	exact
the	دَا	TP	دَا	exact
this	دِس	TP	دِس	exact
truth	تُرُوث	T	تُرُوث	letters
universal	يُونِيفَرَسَل	T	يُونِيفَرَسَل	differs
utopianism	يُتُوبِيَانِيزَم	TP	يُتُوبِيَانِيزَم	differs
with	وَد	T	وَد	exact

Appendix C

Legacy Sources

Earlier files are preserved unmodified in the project.

- `toarabic.ahk`, an AutoHotkey layout from the author’s repository. It assigns one Arabic letter to each Latin key, so `p` and `v` map to key assignments and not to claims about how the sounds are written.
- `pronouncing_translit_chat_version.py`, the pronouncing-based script as pasted from a chat. Its defects are that the vowel table mixes letters and diacritics, the entry for `sh` is never reached, the voiced-*v* test indexes letters with a phoneme index, and no vowel or sukoon placement is done.
- `cli_main_as_pasted.py`, the command-line wrapper, which imports `transliterate_text` from a module named `transliterator`. A shim of that name forwards to the package.

C.1 The old version

The author’s earlier program (the files `old_version_transliterator.py` and `old_version_mappings.py`) converts spelling to the first CMU pronunciation and then writes one letter and one mark per phoneme, with nine fixed words. Its consonant choices are $p \rightarrow \text{ب}$, $v \rightarrow \text{ف}$, hard $g \rightarrow \text{ج}$, $ch \rightarrow \text{ش}$, $ng \rightarrow \text{ن}$ and $zh \rightarrow \text{ز}$. It merges p/b , v/f , hard g and j , ch and sh , and ng and n . It writes each vowel letter before its mark (so `AY` gives `آي`), uses no sukoon outside the fixed words, and takes the first dictionary pronunciation without regard to context. Unrecognised phonemes are dropped silently. The author describes it as not definitive.

The current toolkit differs in three ways that matter. It places the vowel mark on the preceding consonant and writes glides with a sukoon (سَائِي), which is the form on the handwritten page. It writes hard g as غ and ch as ش by default, with switches (`--g jeem`) to return to the old choices. And the v choice is a switch, with ف the default as in the old version, since the author regards v as a voiced f ; *individual* and *universal* in the typed passage use ب and are reproduced by `--v b`. Both versions use only standard Arabic letters, and the author accepts near equivalents such as ب for p when exact recovery of short words is not possible.